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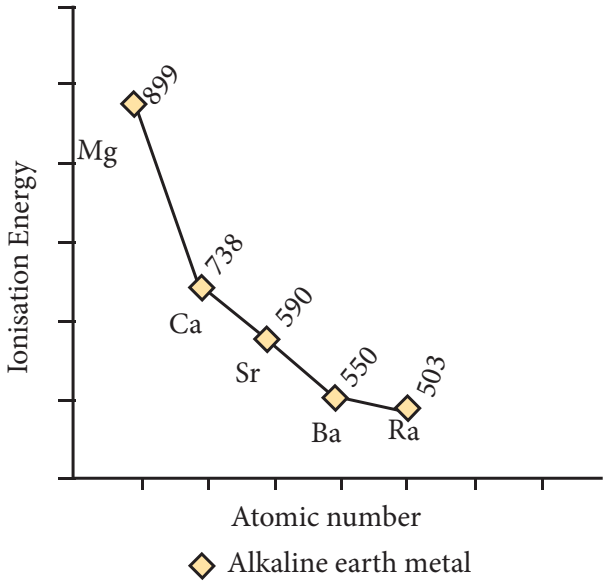

Figure 5.6 Variation of ionisation energy - Alkaline earth metals.

Although IE_1 values of alkaline earth metals are higher than that of alkali metals, the IE_2 values of alkaline earth metals are much smaller than those of alkali metals. This occurs because in alkali metals the second electron is to be removed from a cation, which has already acquired a noble gas configuration. In the case of alkaline earth metals, the second electron is to be removed from a monovalent cation, which still has one electron in the outermost shell. Thus, the second electron can be removed more easily in the case of group 2 elements than in group 1 elements.

Hydration Enthalpies

Compounds of alkaline earth metals are more extensively hydrated than those of alkali metals, because the hydration enthalpies of alkaline earth metal ions are larger than those of alkali metal ions. Like alkali metal ions, the hydration enthalpies of alkaline earth metal ions also decrease with increase in ionic size down the group.

$$Be > Mg > Ca > Sr > Ba$$

e.g., $MgCl_2$ and $CaCl_2$ exist as $MgCl_2.6H_2O$ and $CaCl_2 \cdot 6H_2O$ while NaCl and KCl do not form such hydrates.

Electronegativity

In alkaline earth metals the electronegativity values decrease as we go down the group as seen in the alkali metals.

Flame colour and the spectra:

When the alkaline earth metal salts moistened with concentrated hydrochloric acid are heated on a platinum wire in a flame, they show characteristic coloured flame as shown below.

Table 5.10 Flame Colour and wavelength

Element	Colour	Wavelength (nm)
Calcium	Brick - Red	622
Strontium	Crimson	689
Barium	Apple Green	554

The heat in the flame excites the valence electron to a higher energy level. when it drops back to its actual energy level, the excess energy is emitted as light, whose wavelength is in the visible region as shown in the above table.